SIMATS ENGINEERING

ASSESSMENTTOOL3:TechnicalPresentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloudcomputing technologies including hardware-software evolution, virtualization, web services,and servicemodels to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I_____ (Name / Reg No) certify that this submission is myoriginal work and that I have adhered to the guidelines specified for this assessment. Iunderstand thatany violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Technical Presentation Topics

Preparea6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance,and one practical example.

Here are 10 additional technicalpresentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - o Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - o Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - o Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - o Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - o Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - o Analyze deployment models, business benefits, challenges, and practical examples.

7. **The Role of ArtificialIntelligenceandMachine LearninginCloudPlatforms**
    - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and ContinuousIntegration/Continuous Deployment(CI/CD)inCloudComputing**
    - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **CloudStorageTechnologies:Object,Block,and FileStorageCompared**
    - Compare storage architectures, use cases, and industry adoption patterns.
10. **InternetofThings(IoT)andCloudComputing: BuildingConnectedEcosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **CloudMigrationStrategies:MovingEnterprise ApplicationstotheCloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big DataAnalyticsinCloudComputing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **ServerVirtualizationandItsRoleinCloudComputing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. **Cloud Security: Threats and Protection Mechanisms**

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

    Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. **Software as a Service (SaaS): Architecture and Business Applications**

    Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title
- Aim and objective
- Core technical explanation
- Architecture or process diagram
- Methodology
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria Technical	**Weightage** 30%	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Content		Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Contentis inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Topic:

Introduction

Cloud computing provides computing services over the internet.
It reduces the need for physical hardware.
IaaS stands for Infrastructure as a Service.
PaaS stands for Platform as a Service.
Both models provide flexible and scalable cloud services.

Objective

To understand IaaS and PaaS.
To identify their main features.
To compare their differences.
To understand their advantages and limitations.
To study their real-world applications.

Core technical explanation:

IaaS — Infrastructure as a Service
Provides virtual machines, servers, storage, and networking.
Allows users to manage operating systems.
Offers greater control over infrastructure.
Supports flexible and scalable resources.
Examples: Amazon EC2, Azure Virtual Machines, and Google Compute Engine.
PaaS — Platform as a Service
Provides a platform for application development.
The provider manages servers and operating systems.
Offers development tools and runtime environments.
Simplifies application testing and deployment.
Examples: Google App Engine, Heroku, and Azure App Service.

Architecture or process diagram:

IaaS: Physical servers → Virtual machines → Operating system → Applications.
PaaS: Physical servers → Operating system → Development platform → Applications.
IaaS users manage more of the software stack.
PaaS providers manage most of the underlying infrastructure.
Both models deliver services through the cloud.

Real-world use case:

IaaS
Hosting websites and web servers.
Running virtual machines.
Storing and backing up data.
Supporting disaster recovery.
Running AI and machine-learning workloads.
PaaS
Developing mobile applications.
Building websites.
Creating online shopping applications.
Developing APIs and web services.
Testing and deploying software. text

Advantages and Limitations.

IaaS

Provides greater control.

Offers flexible scalability.

Reduces hardware investment.

Requires technical knowledge.

Needs operating system and security management.

PaaS

Speeds up application development.

Reduces infrastructure management.

Provides built-in development tools.

Offers less infrastructure control.

May create dependence on a cloud provider.

Photo presentation



Final Conclusion

IaaS and PaaS are important cloud computing models.

IaaS offers infrastructure-level flexibility.

PaaS simplifies application development.

Both help save time and resources.

The right choice depends on the user's needs and technical requirements.